

ARRHAT NAG

✉ arrhatnag1@gmail.com  [linkedin.com/in/arrhatnag](https://www.linkedin.com/in/arrhatnag)  github.com/samudragupto  leetcode.com/u/Arrhat

Education

Dr. M.G.R. Educational and Research Institute

Chennai, Tamil Nadu, India

Bachelor of Technology in Computer Science and Engineering

2024 – 2028

Technical Skills

Languages: Python, C++, JavaScript, C, x86 Assembly (NASM), SQL, HTML/CSS, Embedded C, PowerShell

Frameworks & Libraries: Flask, React.js, Node.js, Socket.IO, Streamlit, NumPy, Pandas, Matplotlib, Seaborn, Plotly, scikit-learn, TensorFlow, PyTorch, XGBoost, Chart.js, bcrypt

Developer Tools: Git, GitHub, GitHub Actions, Docker, VS Code, QEMU, MongoDB Atlas, PostgreSQL, SQLite, Arduino IDE, npm

Technologies: REST APIs, WebSocket, CI/CD, Generative AI (Google Gemini), Computer Vision, OCR, IoT, Machine Learning, Systems Programming, Firmware Security, Malware Analysis, Low-Level Programming, Embedded Systems, Data Structures & Algorithms, DBMS

Experience

The Institution of Engineers (India)

Apr 2025 – Present

Joint Secretary Elect

Chennai, Tamil Nadu, India

- Elected as Joint Secretary for IEI Student Chapter at Dr. M.G.R. Educational and Research Institute
- Organize technical workshops, seminars, and events to promote engineering excellence among 500+ students
- Collaborate with faculty and industry professionals to facilitate knowledge-sharing sessions and skill development

Projects

Firmware Implant Detector

Jan 2026 – Present

- **Technologies:** x86 Assembly (NASM), Systems Programming, Malware Analysis, Computer Architecture
- Engineered bare-metal forensic security tool in x86 Assembly that boots directly from USB to audit firmware integrity without OS, eliminating kernel-level rootkit interception risks
- Developed custom multi-stage MBR bootloader with manual CPU mode switching between 16-bit Real Mode and 32-bit Protected Mode, and configured Global Descriptor Table (GDT) for memory segmentation
- Implemented forensic scanning engine featuring CRC32 integrity verification, signature scanning for known implants (LoJax, MosaicRegressor), and entropy analysis heuristics for detecting packed/encrypted malware
- Built hardware abstraction layer interfacing with Interrupt Vector Table (IVT), Option ROMs, and BIOS Data Area (BDA) to detect hooks and anomalies; wrote raw disk I/O driver for cryptographic baseline persistence
- **GitHub:** github.com/samudragupto/Firmware-Detector

NutriScan: AI-Powered Grocery Analytics Platform

Jan 2026 – Present

- **Technologies:** Python, Streamlit, Google Gemini, Docker, GitHub Actions, SQLite, Plotly, Pandas, REST APIs, bcrypt
- Designed full-stack SaaS application auditing grocery receipts for nutritional quality and financial efficiency using Generative AI and Computer Vision with OCR engine for receipt digitization
- Integrated Google Gemini (gemini-3-flash-preview) as real-time nutritional consultant generating context-aware dietary advice and recipes based on user purchasing history
- Engineered CI/CD pipeline using GitHub Actions for automated testing/linting; containerized with Docker for environment consistency across development and production
- Built hybrid data layer combining real-time REST API fetching (OpenFoodFacts, USDA) with SQLite for authentication; developed responsive UI with interactive Plotly dashboards
- **GitHub:** github.com/samudragupto/grocery_health_analyzer

AI-Enabled Defence Cyber Incident & Safety Portal

Nov 2025 – Present

- **Technologies:** Python, Flask, TensorFlow, scikit-learn, MongoDB, React.js, HTML, CSS, JavaScript, REST APIs
- Built AI-powered threat detection platform for defence personnel with ML models analyzing suspicious URLs, images, audio, video, and documents in real-time to identify malicious content and targeted attack patterns
- Implemented multi-factor authentication (smart card + biometric + TOTP) and quantum-resistant encryption with five-layer security architecture meeting ISO 27001, NIST, and FIPS 140-2 compliance standards
- Developed role-based dashboards providing tailored intelligence for service members and CERT-Army analysts with real-time threat maps, confidence scoring, and automated mitigation playbooks
- Integrated global threat feeds with incident correlation to identify emerging attack campaigns targeting defence communities

Patent

System and Method of Portable Electromechanical Braille Display for Visually Impaired Users

Indian Patent Application No. 202541107021 A — Filed: 05/11/2025 — Published: 28/11/2025

Inventors: Arrhat Nag

Designed portable electromechanical Braille display system converting digital text to tactile output using solenoid actuator array in 2x3 configuration, processing unit with Braille mapping logic, actuation control circuit, and power management module enabling low-cost, energy-efficient, modular accessibility solution without piezoelectric actuators.

Publications

Water Footprint - Analysis and Reduction Strategies

Published on Zenodo, Indexed in OpenAIRE (Sep 2025)

DOI: 10.5281/zenodo.17234736

Certifications

IBM TechXchange Dev Day: Open Source LLMs — Virtual Agents — BeMyApp (Jan 2025, Apr 2025)

Python Programming: Upskillist - Multiple Module Certifications (Nov 2024, Apr 2025)

Prompt Engineering for Everyone: IBM Cognitive Class (Nov 2024)

Python 101 for Data Science: IBM Cognitive Class (Nov 2024)

Achievements

Competitive Programming: Active on LeetCode solving 100+ problems in Data Structures and Algorithms

Hackathon Experience: Prototyped real-time IoT + ML system for water-borne disease outbreak prediction

Open Source Contributions: Maintained multiple GitHub repositories with many projects in full-stack, AI/ML etc.